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Affordances and classification: On the significance of a sidebar in James Gibson's last book

Rob Withagen and Anthony Chemero

This article is about a sidebar in James Gibson's last book, The ecological approach to visual perception. In this sidebar, Gibson, the founder of the ecological perspective of perception and action, argued that to perceive an affordance is not to classify an object. Although this sidebar has received scant attention, it is of great significance both historically and for recent discussions about specificity, direct perception, and the functions of the dorsal and ventral streams. It is argued that Gibson's acknowledgment of Wittgenstein's ideas of classification suggests a limited scope of his theory of direct perception—it cannot account for the classification of objects. The implications for both the specification debate and theorizing about the brain's dorsal and ventral pathways are explored. Based on a recent ecological conception of information and direct perception, we ultimately argue that both affordance perception and classification are direct.

Keywords: Affordances; Classification; Direct Perception; Ecological Psychology; Information

1. Introduction

This article is about a sidebar—that is, a boxed comment set off from the main text—in Gibson's (1979/1986) *The ecological approach to visual perception*. In chapter 8 of this book, Gibson described his famous notion of affordances, action possibilities the environment offers an animal. Gibson asserted that the environment of an animal consists of action possibilities and these action possibilities are the primary objects of perception. In a sidebar in that chapter Gibson argued that to perceive an affordance is not to classify an object. This paper is about that sidebar.

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Before we proceed to the main discussion, a few words on the rationale of this paper might be in order. After all, one might wonder how a few lines added at the bottom of a page could justify an entire journal article, especially since these lines have received very little attention in the literature. Although a few authors made a passing reference to the sidebar's central message (e.g., Norman, 2002; Young, 2006), as far as we know only Heft (2001) devoted serious attention to it. We believe that this sidebar is significant in several respects. It is of interest historically in that it reveals something about Gibson's ideas about the scope of his theory of direct perception. It is also of interest for the current debates about the specificity of patterns in the ambient array, the nature of direct perception, and theorizing about the brain's dorsal and the ventral stream. Hence, there is ample reason to devote attention to this sidebar.

The sidebar in question reads as follows:

To perceive an affordance is not to classify an object.

The fact that a stone is a missile does not imply that it cannot be other things as well. It can be a paperweight, a bookend, a hammer, or a pendulum bob. It can be piled on another rock to make a cairn or a stone wall. These affordances are all consistent with one another. The differences between them are not clear-cut, and the arbitrary names by which they are called do not count for perception. If you know what can be done with a graspable detached object, what it can be used for, you can call it whatever you please.

The theory of affordances rescues us from the philosophical muddle of assuming fixed classes of objects, each defined by its common features and then given a name. As Ludwig Wittgenstein knew, you *cannot* specify the necessary and sufficient features of the class of things to which a name is given. They have only a 'family resemblance'. But this does not mean you cannot learn how to use things and perceive their uses. You do not have to classify and label things in order to perceive what they afford. (Gibson, 1979/1986, p. 134)

To explain the significance of these lines, we begin by describing Gibson's revolutionary theory of direct perception. After a brief sketch of the central concepts of "ambient array," "affordances," "specificity," and "direct perception," we will discuss the historical import of the above sidebar. Gibson's reference to Wittgenstein's insights is interesting in that it has implications for the scope of his theory of direct perception—it suggests that Gibson believed that affordances can be directly perceived, but classifying objects is not the result of the detection of specifying patterns in the ambient array. We will discuss this point in its relation to two current debates concerning direct perception: that over the place of direct perception in the two neural pathways in vision, and that over the role of specificity in direct perception. Regarding the former, we will argue that Norman (2002) incorrectly relegates direct perception to the dorsal stream: visual perception involving both the dorsal and ventral streams can be direct. Regarding the latter, we will argue that, given Gibson's conception of the relationship among specificity, information, and direct perception, classification cannot be based on specifying patterns in the array. However, for empirical and conceptual reasons, the

appropriateness of Gibson's conception of this relationship has recently been a matter of debate among ecological psychologists (see e.g., Chemero, 2003, 2009; Fajen, 2005; Jacobs & Michaels, 2002; Withagen, 2004; Withagen & Chemero, 2009). We will argue that according to a recent revision of Gibson's ideas concerning information and direct perception by Withagen and Chemero (2009), both perception of affordances and classification of objects can be direct.

2. Gibson's Theory of Direct Perception

The significance of Gibson's (1966, 1979/1987) view is best understood if you contrast it with the cognitive theory of perception he opposed. Broadly speaking, the traditional, cognitive view holds that perceivers are in only mediated perceptual contact with the environment (see, e.g., Neisser, 1967). This is because the stimulus that impinges upon the senses is impoverished in that it relates ambiguously to the environment; that is, there is a many-to-many relationship between the stimulus pattern and the environmental state of affairs. This means that different environmental states of affairs can result in the same stimulus patterns, and the same environmental state of affairs can give rise to different stimuli. In other words, based upon stimulation alone, the perceiver is in a state of uncertainty about what is "out there." To reduce or eliminate this uncertainty, the perceiver must add information stored in memory to the information gleaned from the stimulus. Internal computation, or what Chemero (2009) called "mental gymnastics," is needed in the perceptual processes. According to this theory, perception is thus conceived as an internal computational process that produces a mental representation that may or may not correspond to the environmental state of affairs.

In developing his ecological view, Gibson aimed at overturning this theory, some version or other of which had dominated thinking about perception ever since Descartes (Reed, 1982). He argued that perceivers are in direct perceptual contact with the environment. Perception is not a mental state in the head; rather, it is the activity of "keeping in touch with the environment" (Gibson, 1979/1986, p. 239). To explain how animals can be in direct perceptual contact with the environment, Gibson criticized the traditional idea that the stimulus information that is available to the senses is impoverished. Gibson conceded that the stimuli that impinge the receptors of a passive perceiver often relate ambiguously to the environment at any given moment. However, he argued that in perceiving their environments, animals are ceaselessly active—their eyes scan the structures in the light, they move their head, their trunk and the rest of the body to create and obtain information about the surroundings. This action on the part of the perceiver generates information for perception. These stimulus patterns that are available to the active perceiver are not at all impoverished—they specify the environment. This idea that stimulus patterns specify the environment is central to Gibson's ecological view of perception.

According to Gibson, then, specifying patterns are to be found in the ambient energy arrays that surround an active perceiver. The structure of these arrays is the

result of physical properties of the environment, the (changing) relation between animal and environment, and physical laws (e.g., laws of optics and acoustics; Gibson 1979/1986, pp. 86–92). In their exposition of the laws of optics, Michaels and Carello (1981, p. 22) explain how different environmental objects reflect the light in different ways. For example, because of its physical properties, a lump of coal reflects light differently than a cardboard box. Because patterns in the array are generated lawfully, they can contain a specification of the environment. That is, the patterns in the optic array relate one-to-one to the environmental properties that have structured the light. The availability of these specifying patterns, Gibson argued, allows for a direct perceptual contact with the environment; because the patterns are lawfully related to properties of the environment, they carry information about those environmental properties. This allows animals to perceive these properties simply by detecting the patterns. Moreover, there is no need for mental gymnastics: the patterns in the optic array carry sufficient information to allow perception of the environmental properties without internal computations. Animals are simply in direct perceptual contact with their environment by picking-up specifying patterns in the ambient array.

Gibson not only developed an alternative theory of how animals perceive, but also of what animals perceive. The latter is Gibson's famous theory of affordances. Affordances are the action possibilities the environment offers the animal. For example, a cup affords grasping; a chair, a stone, and a couch might afford sitting on; water affords drinking; and so on. Gibson asserted that animals perceive their environments primarily in terms of affordances. There are several aspects of the concept of affordances that are relevant for the purposes of the paper. First, affordances are determined by the physical properties of the environment relative to the action capabilities of the animal. This means that the same object can afford different behaviors to different animals. As an example, a cup might afford grasping for an adult human, but not for an infant or a spider. Second, an environment consisting of affordances is a meaningful environment. Affordances determine what an animal can do in a certain setting and, thus, what the setting means to the animal. Hence, if there are patterns available in the ambient arrays that specify affordances, animals can directly perceive meaning. That is, meaning is not added to the stimulus by the animal in an internal mental process; rather, it is discovered during an animal's exploration of the environment.

3. The Relevance of the Sidebar

With this broad sketch of Gibson's ecological perspective, we can understand the significance of the sidebar this paper is about. In the sidebar Gibson made several claims.¹ First, he distinguished the classification of objects from the perception of affordances. The title of the sidebar reads: "To perceive an affordance is not to classify an object," and he ended the sidebar by claiming that, "you do not have to classify and label things in order to perceive what they afford" (Gibson, 1979/1986,

p. 134). In other words, the perception of affordances is independent of the classification of objects. This distinction recurs in recent discussions on the functioning of the dorsal and ventral streams in vision. For example, Norman (2002) allied these two visual streams with the two different approaches to vision. In his view, dorsal stream vision is concerned with the perception of affordances and the online guidance of action, and can be studied using Gibsonian ecological psychology. The ventral stream is involved with classifying and representing the environment, and can be studied using the methods of computational cognitive neuroscience. In making this division, Norman is following Gibson's distinction between the perception of affordances and the classification of objects.

Another important aspect of the sidebar is that Gibson argued that an individual object can afford multiple behaviors. The stone affords throwing, carrying, rolling, standing upon, and so on. This claim, which Gibson had also made at other points in his work, has provoked some debate. Cutting (1982, p. 216) for instance argued that the fact that a single object affords myriad behaviors implies an inconsistency in Gibson's ecological perspective. On Cutting's reading, Gibson claimed that affordances constrain perception and action. However, if a single object affords myriad behaviors, then affordances, in and of themselves, cannot sufficiently constrain the animal behavior. Heft (1989), on the other hand, noted that the multiplicity of affordances in a single object is fully consistent with the theme of intentionality that is central to the ecological approach. As he put it, "the affordances of an object are realized in relation to some intentional act in the individual's behavior repertoire" (Heft, 1989, p. 21).

Lastly, and what is perhaps of greatest interest in the sidebar, is the reference to Wittgenstein's writings on the classification of objects. Gibson argued that the theory of affordances "rescues us from the philosophical muddle of assuming fixed classes of objects" (p. 134) and referred to the later work of Wittgenstein. In his *Philosophische Untersuchungen* [*Philosophical Investigations*], Wittgenstein (1953) argued that there are no determinate boundaries of classes:

Consider for example the proceedings that we call 'games'. I mean board-games, card-games, ball-games, Olympic games, and so on. What is common to them all?—Don't say: "there *must* be something common, or they would not be called 'games'"—but *look* and *see* whether there is anything common to all.—For if you look at them you will not see something that is common to all, but similarities, relationships, and a whole series of them at that. To repeat: don't think, but look! (section 66)

According to Wittgenstein, this is true not only for the class 'games' but for all classes. For example, the objects that a human perceives as a chair do not have any one physical property in common. Wittgenstein introduced the concept of "family resemblance" to denote the similarities of certain objects that make them members of the same class. That is, the members of a class are recognizably similar, although they do not all share any single property. Gibson was clearly aware of Wittgenstein's insights regarding classes. He wrote in the sidebar, "as Ludwig Wittgenstein knew, you *cannot* specify the necessary and sufficient features of the class of things to which

a name is given” (p. 134). This assertion is interesting because it has implications for the scope of Gibson’s theory of direct perception. Although Gibson did not explicitly state this conclusion himself in the sidebar, Wittgenstein’s insights imply that class memberships are not specified by patterns in the array or flow. Recall, that Gibson claimed that the pattern in the optic arrays and flows are determined by physical properties of the environment, the laws of optics, and the (changing) animal-environment relation. This means that if classes do not have determinate boundaries and, thus, do not have a physical property in common, then there are no patterns in the ambient light that specify class membership.

Gibson suggested that for affordances things are different. Indeed, the “theory of affordances rescues us” (Gibson, 1979/1986, p. 134) from this muddle in that the affordances of the environment for a particular animal have determinate boundaries. Because affordances are relations between the physical properties of the world and the action capabilities of the body (e.g., Chemero, 2009; Heft, 1989; Michaels, 2003; Stoffregen, 2000), these action capabilities determine what physical properties an object must have to afford a certain behavior (see also Heft, 2001, p. 130). In the literature on affordances these determinate boundaries are often termed ‘critical points’ (e.g., Warren, 1984) or ‘affordance thresholds’ (e.g., Ishak, Adolph, & Lin, 2008), and they determine the environmental circumstances in which affordances are present. For example, all the objects that are graspable with the right hand for a particular human have certain physical properties in common: they cannot be bigger than the span of that person’s hand, they must be not too cold or too hot, their surfaces must not be sharp, and so on. The abilities of the body set the definite boundaries of the class of objects that afford a certain behavior to an animal, allowing the affordance to be specified by patterns in the ambient arrays. Indeed, ever since Warren’s (1984) pioneering study of affordances, experimental work (e.g., Gill, Adolph, & Vereijken, 2009; Mark, 1987; Ishak, Adolph, & Lin, 2008) has focused on determining “critical points,” the environmental boundaries at which possibilities for individual actors change. For example, for an individual newly-walking infant, there is a slope angle at which walking affordances disappear: at slopes less steep than this angle, she will almost always succeed in descending to her waiting mother; at steeper slopes, she is almost always unwilling to descend, and typically falls if coaxed down by her mother (Ishak, Adolph, & Lin, 2008). Critical points are perceivable, determinate boundaries for action possibilities.

The fact that affordances have determinate boundaries, however, does not imply that they are always specified by patterns in the ambient array. For a pattern in the light to carry information specifying an action possibility, the relevant physical properties must determine the pattern and this is not always the case. Gibson talked in this context about “misinformation for affordances” (1979/1986, p. 142). He claimed that, “when Koffka asserted that ‘each thing says what it is’, he failed to mention that it may lie. More exactly, a thing may not look like what it is” (Gibson, 1979/1986, p. 143). An example of such misinformation is the mimicry that occurs frequently in the animate world. Animals that are edible for a certain species sometimes look like inedible animals, and thus have a higher chance of survival and

reproduction. Hence, the pattern in the ambient array is not determined by the relevant physical properties (i.e., the chemical substance that is edible) but by its outward appearance. Hence, although affordances have determinate boundaries and therefore can be specified by patterns in the ambient array, they are not always actually specified. Whether or not they are, depends on whether the relevant properties determine the structure of the ambient array.

3.1. Did Gibson Believe that his Theory of Direct Perception has a Limited Scope?

It has been argued that Gibson was not very specific in what he meant by “direct perception” (Costall, 1989). However, the vast majority of neo-Gibsonians interpret his theory of perception as a claim on the nature of the stimulus information that perceivers use. To them, direct perception means the detection of specifying information (see e.g., Burton & Turvey, 1990; Carello & Turvey, 2004; Michaels & Beek, 1995; Turvey, 1990). We agree that this connection between direct perception and specifying patterns is central to Gibson’s perspective. Indeed, he argued that animals perceive their environment directly *because* they detect specifying information. Hence, the reference to Wittgenstein in the context of classification is very interesting. Although, as mentioned above, Gibson did not explicitly draw the conclusion that direct classification is not possible, we believe that this is implied by the sidebar. After all, he not only claimed that the concept of affordances defeats the idea of fixed classes of objects, but also emphasized that there are no necessary and sufficient conditions for class membership. A corollary of this claim is that class membership is not specified by patterns in the ambient array. Hence, if direct perception is defined as the detection of specifying patterns, then one cannot directly perceive an object as belonging to a class. Direct perception is limited to the perception of (some) affordances. That is, whether something is a member of the class ‘ball’ is not specified in the array and, so, is not directly perceivable; on the other hand, whether something affords throwing by an individual is both specified and directly perceivable.

Of course, there are ways to avoid this conclusion. One might argue that humans do not classify objects, at least not perceptually. The only objects of perception are affordances, rendering the theory of direct perception applicable to all perceptual awareness. The contemporary ecological psychologist Stoffregen (2000) appeared to argue for such a position. Discussing the relation between events and affordances, he suggested that affordances are the sole objects of perception:

I have argued that the essential difference between affordances and events is that events are not defined relative to the animal and so do not, in and of themselves, have consequences for behavior. Given this and assuming that affordances actually are perceived, we are left with the question of why, in an ecological theory, animals would perceive events. (Stoffregen, 2000, p. 18)

Stoffregen’s argument was intended to criticize ecological studies of event perception, but the same reasoning might be applied to perception of objects-as-members-of-classes. Animals, one might argue, also do not perceive objects as belonging to a class, because, in and of themselves, objects-as-members-of-classes

also have no consequences for the animal's behavior. However, as Stoffregen (2000) himself acknowledged, Gibson did not defend such a radical position. In the specific sidebar that we write about he claimed that the perception of affordances does not require classification, but Gibson did not deny that human beings perceive objects as belonging to classes. Indeed, although the introduction of the concept of affordances represents a landmark in the development of his ecological view (Heft, 2001; Lombardo, 1987; Reed, 1988), it was not accompanied by a radical claim that they are the only objects of perception. In fact, even in his last book, Gibson (1979/1986) described the animal's environment not only in terms of affordances, but also in terms of surfaces, events, and objects. Hence, on our reading Gibson recognized that humans can also perceive objects as belonging to particular classes. However, the acknowledgment of Wittgenstein's ideas of "family resemblance" in the above sidebar suggests that Gibson believed that class memberships are not specified by patterns in the ambient arrays. This means that Gibson's theory of direct perception would be limited to the perception of affordances.

4. Implications for the Debate on Specificity

Gibson's sidebar is also relevant for the recent debates on the specificity of patterns in ambient arrays, at least if one recognizes that humans classify objects on the basis of perceptual variables. It is our contention that human beings do so. That is, they directly perceive chairs, tables, bicycles, couches, paintings, and many objects as such.

In the literature, however, it has been suggested that the classification of objects is not the result of the pick-up of perceptual variables; rather, classification is a cognitive activity that depends upon, but differs from, the direct perception of affordances. For example, in discussing the sidebar that this paper is about, Heft (2001) asserts that classification is based on the direct perception of affordances. Following William James, he claimed that "categorization is the work of second-order processes; it is a thinking about immediate experience" (Heft, 2001, p. 130). Neisser (1987) seems to defend a similar position. He argues that "perception is based on the direct pick-up of objectively existing information, while [classification] goes beyond that information on the basis of beliefs about world" (Neisser, 1987, p. 4). However, although we believe that (the development of) categorization is largely determined by the perceiving-acting body (Johnson, 1990; Lakoff & Johnson, 1999; Sheets-Johnstone, 1999), we think it is not a second-order process based on the direct perception of affordances. There are several reasons for this. First, classes are only very loosely related to affordances and certainly cannot be defined in terms of them. For example, chairs are not always objects one can sit on; and not all objects one can sit on are chairs. Assuming that classes can be defined in terms of affordances is not taking into account Wittgenstein's insights about the indeterminacy of the boundaries of classes, which Gibson acknowledged. Second, and more importantly, recent developments in neuroscience suggest that classification and the perception of affordances are largely subserved by different visual streams in the brain

(e.g., Norman, 2002; see below). And although there is and should be crosstalk between these different streams (Young, 2006), the functioning of one stream is not based on the other. Instead, they work relatively independently, each driven by stimulus information. This suggests that both classification and affordance perception are based on perceptual variables.

If one accepts this claim, then Gibson's reference to Wittgenstein's ideas has serious consequences for the recent debates on specificity in the ecological literature. Traditionally, ecological psychologists have searched for specifying patterns in the ambient array (see e.g., Burton & Turvey, 1990; Carello & Turvey, 2004; Turvey & Carello, 1995; Turvey, 1990). Over the last several decades, there has been discussion about what allows energy patterns to be specific to the environment (e.g., Runeson, 1988) and whether such patterns are to be found in the individual, energy-specific arrays or only in the global energy array (e.g., Stoffregen & Bardy, 2001). It is important to note that although this search for specificity is central to the ecological movement, some ecological psychologists have recognized that perceivers in some circumstances also rely on non-specifying variables (for reviews, see Fajen, 2005; Jacobs & Michaels, 2002; Withagen & Chemero, 2009). In line with Gibson's ideas on misinformation, some authors have argued that perhaps not all perceivable environmental properties are specified by patterns in the ambient array (Chemero, 2009; Jacobs, 2001, p. 199; Turvey & Carello, 1995, p. 483). Moreover, empirical studies of perceptual learning have shown that novice perceivers tend to rely on non-specifying variables (Jacobs, Michaels, & Runeson, 2000; Michaels & de Vries, 1998; Runeson, Juslin, & Olsson, 2000).

However, if class membership is not specified by patterns in the ambient array, and in our daily activity we classify objects on the basis of perceptual variables, then the detection of non-specifying variables occurs on a larger scale than has been assumed in ecological psychology. It occurs not only in exceptional circumstances (i.e. when affordances are not specified by patterns or in case of a novice perceiver), but is common in daily life. This provides a serious challenge for the ecological theory of perception. Theoretically speaking there are two options. First, one might develop the ecological theory of perception such that it can deal with perceiving the environment in terms of classes. Second, one might claim that the ecological theory can explain only the perception of affordances, and not the classification of objects. Some have taken recent discussions on the ventral and dorsal streams of vision as recommending the second of these options.

5. The Dorsal Stream, Ventral Stream, and Direct Perception

In discussing the functions of the dorsal and ventral stream, Norman (2002) and Young (2006) have referred to Gibson's idea that to perceive an affordance is not to classify an object. Gibson's sidebar is indeed relevant to this discussion. If direct perception requires the detection of specifying variables, and classes are not specified but affordances are, then one might argue that there are two modes of

perception: one direct mode (concerned with affordances) and one indirect mode (concerned with classification). As mentioned above, this is exactly the position that Norman (2002) developed in his theory of the brain's dorsal and ventral visual streams. Inspired by Milner and Goodale's (1995) two visual streams model, he argued that there are two modes of perception: a direct, ecological one and a constructivist, cognitive one. The direct one is subserved by the dorsal stream and is concerned with the perception of affordances and the online guidance of action. The constructivist mode is concerned with the recognition and classification of objects and is subserved by the ventral stream. With this theory, Norman tried to reconcile two different approaches to perception that have been considered incompatible with each other.

Several authors, however, have argued that Norman's distinction of a dorsal, ecological stream and a ventral, cognitive stream is too simplistic (see the extensive commentaries on his target article). For the purposes of the present paper, Young's (2006) comment is of special interest. He argues quite convincingly that affordance perception is not subserved by the dorsal stream alone. Some affordance perception is the result of ventral stream activity (see also Postma, van der Lubbe, & Zuidhoek, 2002). Citing the well-documented inabilities of neuropsychological patients discussed in the literature, Young pointed out that patients with a ventral lesion but an intact dorsal stream are not capable of perceiving some affordances. The well-known patient DF, for example, is still capable of grasping a pair of scissors but has severe difficulty grasping it at the appropriate place (the handles), revealing no understanding of the proper function of the object (Carey, Harvey, & Milner, 1996). Patients with a properly working ventral stream but a damaged dorsal stream, however, can do this. IG for example can recognize a hammer and describe and demonstrate its function. However, she has difficulty with locating and grasping it in real-time (Milner et al., 2001). This suggests that the perception of some affordances is subserved by the dorsal stream, whereas the perception of others depends on a properly working ventral stream.

The fact that the ventral pathway is also involved in the perception of affordances raises several questions: is some affordance perception indirect (assuming that the ventral stream is constructivist)? Or does the ventral stream perceive both directly and indirectly (directly in the case of affordances and indirectly in the case of classification)? Neither of these options is very elegant or parsimonious. Instead, we prefer to follow those who believe that, in principle, any kind of perception might be direct (e.g., Michaels, 2000; Michaels, Withagen, Jacobs, Zaal, & Bongers, 2001). However, because class membership is not specified in the array, to account for perceptual classification, the theory of direct perception cannot be based on the principle of specificity. In the remainder of this paper, we sketch the outlines of such a theory.

5.1. Direct Perception and Non-Specificity

The key point to understanding that classification can be direct is a re-conceptualization of information and direct perception so that they are not

defined in terms of specifying patterns in the ambient array (Chemero, 2003, 2009; Withagen, 2004; Withagen & Chemero, 2009). To begin this reconceptualization, it is worth pointing out that Gibson and Gibsonian psychologists are quite unusual in defining information in terms of specification. Their definition amounts to the claim that information about some feature of the environment exists in a pattern in the array only when there is a one-to-one relationship between the pattern and that feature of the environment. Put in terms of probabilities, the claim is that an array pattern A carries information about an environmental feature F if and only if the probability of F, given A, is 1. There are many conceptual and empirical reasons to think that this is too strenuous a requirement on allowing a pattern to carry information about the environment (see the work by Chemero and Withagen cited above, as well as Barwise & Seligman, 1994; Millikan, 2000, for some of these reasons).

Here we will present an example from Withagen and Chemero (2009) that offers a case in point. According to Wikipedia entry on eyeglasses (as of March 4, 2010), the first very clear glass was produced by Abbas Ibn Firnas in the 9th century. Whether or not this is a reliable account of the origin of clear glass, it is certainly true that for the vast majority of evolutionary time, there was no clear glass. Suppose that in all the time before there was clear glass, optic array patterns of type P specified clear flight paths (for insects, birds, flying squirrels, etc.). This is the case because the presence of pattern P was one-to-one related with clear flight paths. The question we wish to ask is, what happened on the day when clear glass was invented? Since P now appears both when there is a clear flight path and when there is a glass sheet, P is no longer one-to-one related with clear flight paths and, therefore, no longer carries information about the presence of clear flight paths on Gibson's view of information as specification. Since direct perception is also defined in terms of specifying information, fliers go, on one day, from perceiving flight paths directly to perceiving them indirectly. We hope that this strikes readers as absurd. We can avoid this absurdity by separating specification from our definition of information and direct perception.

We suggest that a token array pattern *p* can carry information about token environmental feature *f* whenever there is some constraint *C* that holds between array patterns of type *P* and environmental features of type *F* (Chemero, 2009). Importantly, constraints need not guarantee the presence of some feature, given the presence of some pattern. That is, *C* could be a natural law, but it could also be a custom or other regularity. This understanding of information allows patterns to carry information both when the constraint is exceptionless and when it is not, which is to say that it allows a particular token *p* to carry information about some feature *f*, even if *F*s are only sometimes present when *P*s are present. In other words, there is information in optic array patterns that do not specify environmental patterns. To go back to the example of clear flight paths, according to this conception of information, the optic pattern *P* informs about the presence of such paths if the two are reliably correlated. Relying on such patterns will therefore generally result in adaptive perception. However, occasionally the bird's perception will err and he might crash into a window.

Based on this understanding of information, Withagen and Chemero suggest that specification is also not central to direct perception (Chemero, 2003, 2009; Withagen, 2004; Withagen & Chemero, 2009). They took inspiration from the following lines from Gibson: “when I assert that perception of the environment is direct, I mean that it is not mediated by retinal pictures, neural pictures, or mental pictures” (Gibson, 1979/1986, p. 147). Thus, they claim that perception is direct whenever it is not mediated by mental representations. Direct perception is a coupling between the perceiver and its environment, via information in the array. This is the case whether the information is carried by specifying or non-specifying perceptual variables.

Consider an outfielder who tries to catch a fly ball. To be at the right place at the right time, the outfielder has to establish a coupling between her movements and the trajectory of the ball. This coupling is established by detecting patterns in the optic array. Although the strength of the coupling depends on the usefulness of the variable detected, the nature of the perceptual process does not. When relying on a specifying pattern, there is a tight coupling between the trajectory of the ball and the movements of the outfielder, giving rise to a successful catch. When the outfielder relies on a non-specifying variable, on the other hand, there is still a direct coupling between the ball and the outfielder’s movement, but the pattern in the light the outfielder is relying on is only probabilistically correlated with the arrival of the ball. This means that the outfielder sometimes might not be at the right place at the right time and thus might miss the ball. Yet, perceptually the same process is going on. The outfielder detects a pattern in the array and her movement is directly guided by it. There is no need for the outfielder to represent the ball. In both cases, there is a direct coupling of ball and perceiver, established via the detection of optical patterns.

For current purposes, the most important consequence of taking this coupling as the constitution of direct perception is that perception is, by definition, direct. Perception is always a matter of tracking something that is present in the environment. Because animals are coupled to the perceived when they track it, there is never need to call upon representations. Thus regardless of the nature of the informational variable detected (i.e., specifying or non-specifying), perception is nonrepresentational, hence direct. Explaining how we think about our grandmothers or plan vacations might require invoking something like a representation, but perception never does. The question to be answered in the next section is whether perceiving objects as members of classes could be direct in the sense described here, or is it more like thinking about one’s grandmother.

5.2. *Direct Classification*

The purpose of Gibson’s sidebar was to contrast affordance perception and classification. As is embodied in the very notions of affordance threshold and critical point discussed above, action possibilities can be characterized definitively. An object either is or is not graspable by a particular animal. Class membership, on the other hand, is a matter of family resemblance. There is no set of necessary and sufficient conditions on being classifiable as a chair, and there will be many borderline cases of

objects that are kind of chairs, but also kind of non-chairs. Another way to put this point is to say that whether something is a member of the class 'chair' is incompletely determined by its physical features, and so information about whether some object is a chair cannot be carried by a specifying perceptual variable. Hence, according to Gibson's conception of direct perception as the detection of specifying variables, one cannot perceive that an object is a chair directly; it requires mental gymnastics. And according to Norman, visually classifying some object as a chair is an activity of the cognitive, ventral stream. If one adopts our alternative understandings of information and direct perception, however, neither of these conclusions follow.

Among the many reasons that being a member of the class 'chair' is not specified by optical variable is that there are many things that look exactly like chairs but are in fact sculptures, or museum exhibits set up to exemplify a particular design movement. Conversely, there are many chairs that look nothing like chairs, such as the Capitello chair, which looks like the top of a Roman column, or the Joe chair, which looks like an oversized baseball mitt. We have proposed, however, that non-specifying perceptual variables can also carry information, despite the fact that relying on them can sometimes lead to perceptual error. We have also proposed that direct perception is a matter of using information in perceptual variables to maintain a coupling between animal and environment. This allows the claim that classification can also be direct. The idea is that there are perceptual variables that carry information about objects being members of classes, without specifying that those objects are members of those classes. Indeed, although classes do not have determinate boundaries, the similarities and relationships among the members of the class imply that there are patterns in the array that correlate with classes, but are not specific to them. One can classify an object by picking up this information, and one can do so without mental gymnastics. This is to say that direct classification is possible. To continue with our example, one can classify an object as a chair directly by picking up information in perceptual variables. Doing so is not a guarantee of success at classifying it: using non-specifying visual information to directly classify objects as chairs will work most of the time, but also might lead us to occasionally sit on a museum exhibit or to pass by a Capitello chair even when we are looking for a chair.

Our claim that classification can be direct expands the scope of direct perception. This expansion is justifiable only by adopting an understanding of information that differs from Gibson's in virtue of allowing non-specifying perceptual variables to carry information. As mentioned earlier, Gibson had an atypically restrictive an understanding of information, and there are many reasons to think it was too strict (Barwise & Seligman, 1994; Chemero, 2003, 2009; Millikan, 2000; Withagen, 2004; Withagen & Chemero, 2009). This overly restrictive understanding of information means that direct classification is impossible. A more appropriate, broader conception of information, however, implies that one can also perceive objects as members of classes directly.

6. Concluding Remarks

We began this paper by discussing an important but little discussed sidebar in Gibson's *The ecological approach to visual perception*. In this comment, Gibson argued that classifying objects and perceiving affordances are different activities. He cited Wittgenstein's discussion of category members as having a family resemblance, suggesting that he was aware that classes cannot be specified by patterns in the ambient array. Given Gibson's views of specification, information, and direct perception, this implies that perceiving an object as a member of a class is indirect. This is in line with Norman's division of visual perception into the ecological, dorsal stream for perceiving affordances and cognitivist, ventral stream for classifying objects. Contemporary ecological psychologists who follow Gibson's views of information-as-specification should agree with Norman that only some of perception is direct.

In contrast, we have argued that information should be understood independently of specification, which implies that perception can be direct without relying on specifying perceptual variables. This allows substantially more of our perception to be direct, including our perception of objects as belonging to classes. This claim puts us in an unusual position vis-à-vis Gibson's views. On one hand, contemporary ecological psychologists tend to think that information-as-specification is the central tenet of Gibson's ecological approach. Since we reject this view of information, our view does not seem very Gibsonian at all. On the other hand, contemporary mainstream (i.e., non-ecological) psychologists tend to take direct perception to be the most important (and unduly radical) tenet of Gibson's psychology. Since we believe that perception based on non-specifying variables is also direct, our view seems extremely Gibsonian.

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Note

- [1] In comments on an earlier version of this paper, Harry Heft recommended a reading of the sidebar that differs from ours. Heft interprets the sidebar as doing two things. First, by claiming that naming objects is not the same as perceiving, Gibson is emphasizing his claim that perception is direct. Second, Gibson's invocation of Wittgenstein is intended to point out that the fact that an object can have multiple affordances is consistent with Wittgenstein's claim that there are no necessary and sufficient conditions for class membership. We agree that Gibson is doing the first of these in the sidebar, but, as will become clear below, we read the sidebar very differently than Heft does. For one thing, it seems to us that the multiplicity of affordances of single objects is unrelated to classification, and Gibson is not claiming that they are related. Indeed, the title of the sidebar is "To perceive an affordance is not to classify an object," which suggests that Gibson is doing the opposite of what Heft suggests, and pointing out the profound differences between

perception of affordances and classification. The perception of affordances is direct, and based on information that specifies them. Gibson invokes Wittgenstein in the sidebar to say that there are no necessary and sufficient conditions determining class membership, suggesting that there cannot be information specifying what class an object belongs to.

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